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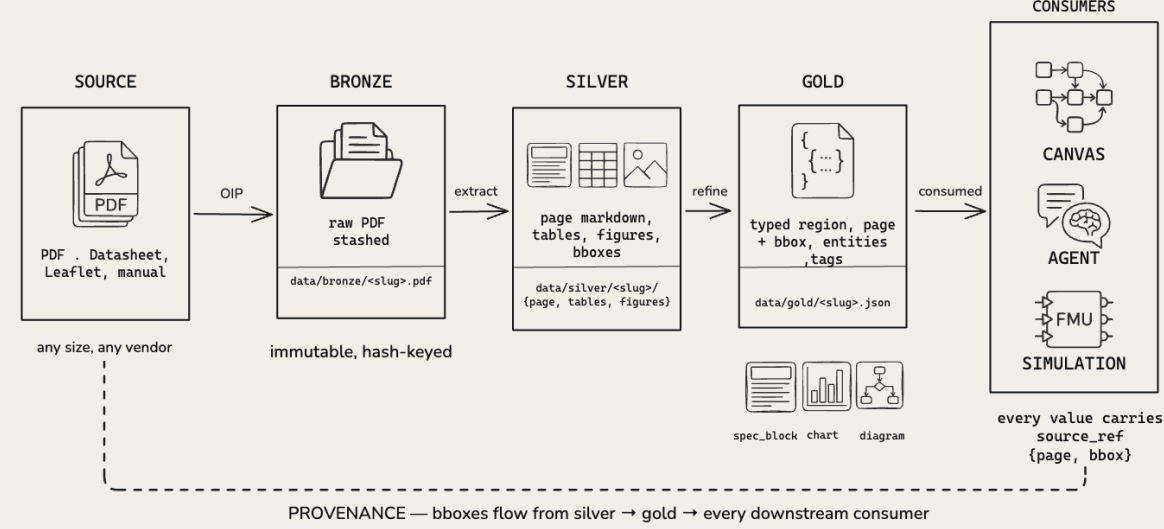
### PROBLEM

AI is already part of engineering work and won't go away. But facts without a clear source can't be trusted, and shouldn't be. Verifying each claim by opening PDFs and scrolling defeats the speed AI was supposed to give.

### WHAT IS ANCHOR

Anchor is a tool for working with engineering documents. From a PDF datasheet, an FMU, a CAD file, or anything else exposed through the Open Ingestion Protocol (OIP), Anchor ingests it and extracts the parts (spec rows, charts, ports, parameters) and makes each one independently addressable by humans and AI agents. Every value keeps a reference back to where it came from in the source. Anchor ships with support for PDF, and FMU ingestion, whereas CAD and SysML extensions are under development.

### INGESTION



A vision-LLM region-extraction step at gold keeps each region semantically coherent: a table header stays with its table, a spec block stays whole. Every value flows forward with its source page and bbox.

### BUILT

- Engineering PDFs go through bronze → silver → gold with a vision-LLM region polish; an FMU extension ships alongside as an example for other source kinds.
- MCP tools cover canvas commands and each producer's actions, namespaced (pdf.\*, fmu.\*, ...) so producers coexist.
- Generic node types (concept, entity, fact, document, spec, area, note) ship built-in; extensions register their own renderers at runtime.
- Pure domain core with HTTP, MCP, and a CLI as peer adapters; import-linter contracts keep the boundaries clean in CI. Vendor SDKs live only in extension infra; the core never sees them.
- JSONL event log per workspace; plain-file artefacts per document. No database, no queue. Multi-client sync (~50 ms) is a side effect of the in-memory bus.

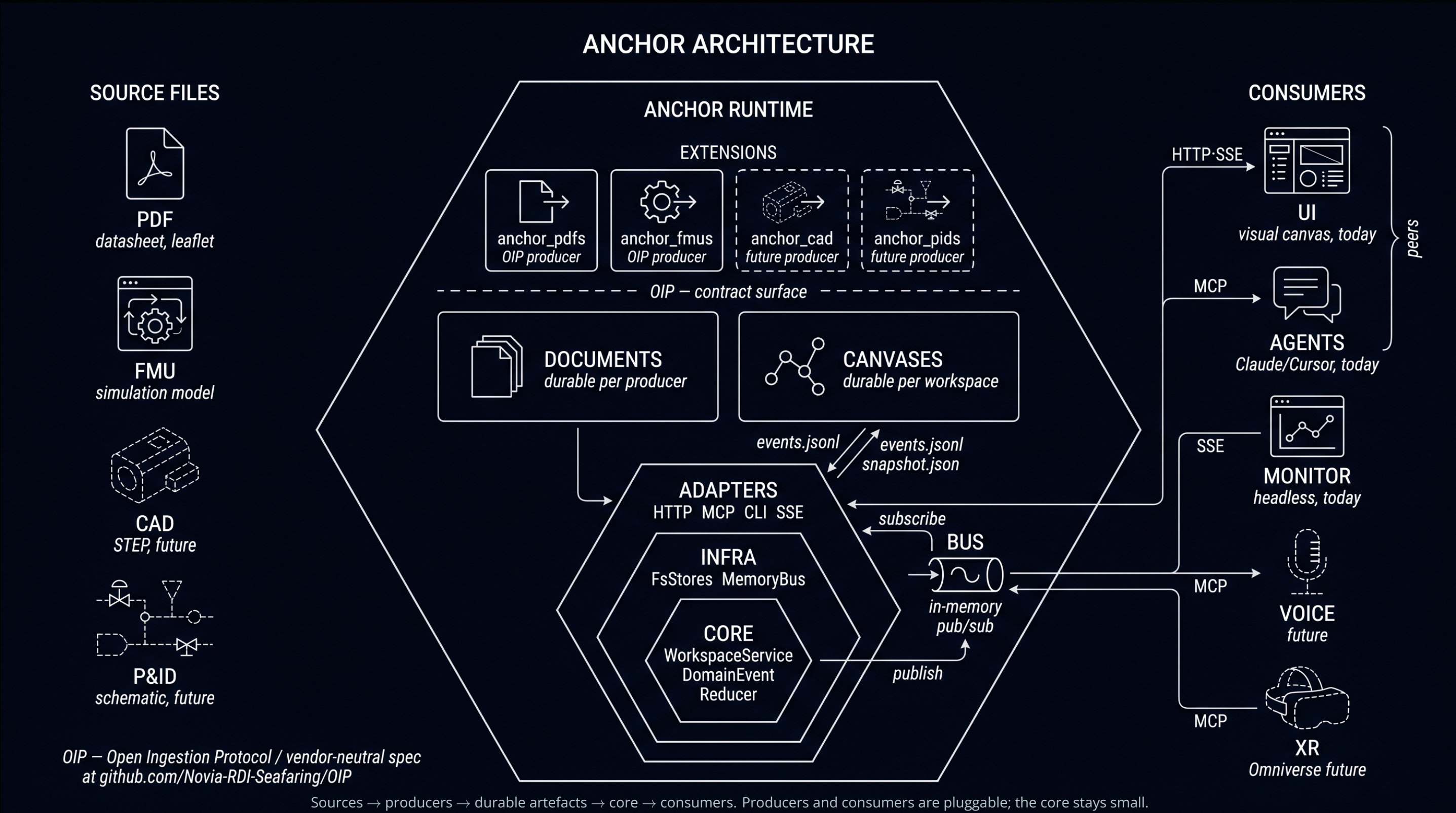
### PORTABLE

A canvas is a folder: events.jsonl plus a derived snapshot. Tar it, copy it to another machine, replay; same state, same provenance. Audit, time-travel, and migration are mechanical, not features that have to be added.

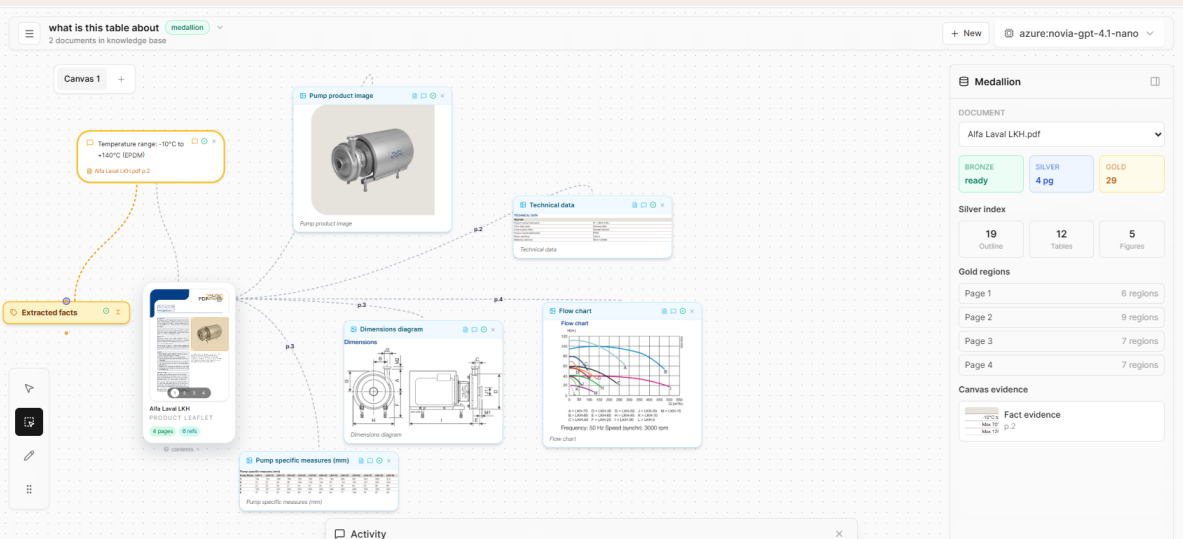
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Every spec your agent quotes  
points back to the  
page it came from.

Datasheet → canvas → simulation.  
Each step keeps the source.

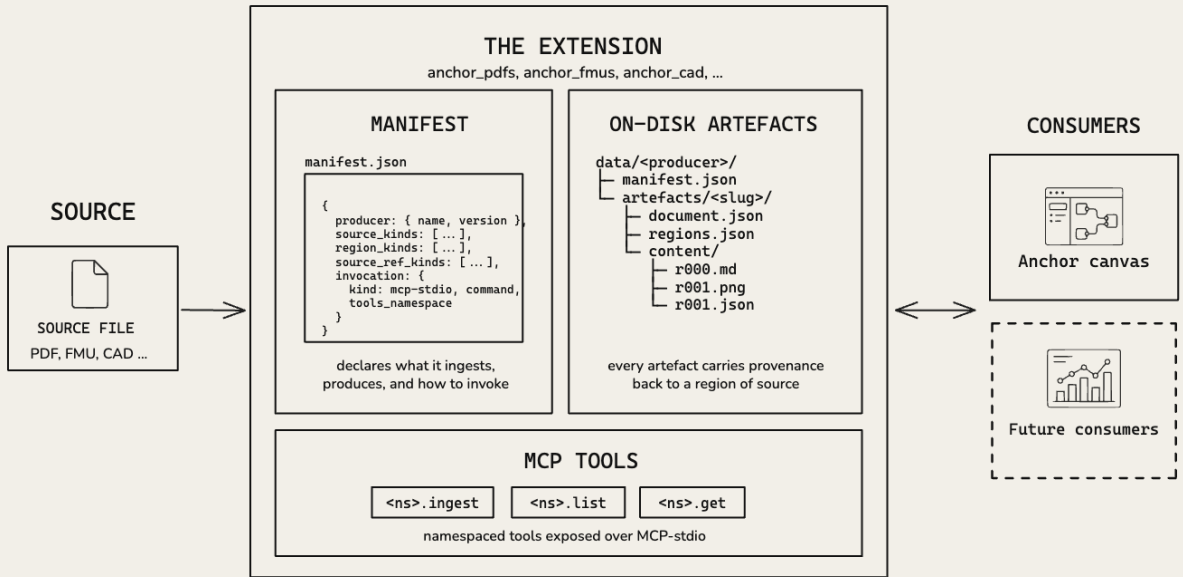


### THE CANVAS



Cards carry source crops with a one-click jump back to the PDF page. Drop a file onto the canvas and the producer that claims its kind ingests it in place.

### EXTENSIONS



An OIP extension is three things: a manifest, on-disk artefacts, and MCP tools. The shipped PDF producer is the working example; anything that satisfies the contract plugs in the same way.

### BRING YOUR OWN HARNESS

The UI, MCP agents, and a CLI share one WorkspaceService and one event bus; humans and agent harnesses drive the canvas through identical paths. New interfaces (voice, XR, a chat monitor) plug in the same way.

### READ MORE



github.com/Novia-RDI-Seafaring/anchor

Source code and local setup instructions.

Industry presentation · Vaasa, 2026